

Review of Similarity and Distance Metrics

Key Concepts

- **Similarity:** Measures how "alike" two objects are (range: $[0, 1]$, 1 = identical).
- **Distance:** Measures how "different" two objects are (range: $[0, \infty)$, 0 = identical).
- Choice depends on: **data type** (vectors, text, sets, binary), **data properties** (sparsity, dimensionality), and **research goal**.

2.1 Core Similarity Metrics

A. Cosine Similarity

- **Definition:** Measures the cosine of the angle between two vectors in a high-dimensional space. Focuses on **direction**, not magnitude.

- **Formula:**
$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{|A||B|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

- **Examples:**
 - Vectors $A = [1, 2, 3]$, $B = [2, 4, 6]$: Cosine Similarity = 1.0 (same direction).
 - Vectors $A = [1, 0, -1]$, $B = [0, 1, 0]$: Cosine Similarity = 0 (perpendicular).
- **Use Cases:** Text classification (TF-IDF vectors), recommendation systems (user/item embeddings), high-dimensional data (e.g., CV feature vectors).
- **Pros:** Robust to feature scaling; ignores magnitude (e.g., "cat" vs "cats" in text).
- **Cons:** Fails if magnitude matters (e.g., comparing user activity counts).

B. Jaccard Similarity (Jaccard Index)

- **Definition:** Measures similarity between two **sets** as the ratio of their intersection to their union.

- **Formula:**
$$\text{Jaccard Similarity}(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

- **Examples:**
 - Sets $A = a, b, c$, $B = b, c, d$: Jaccard = $2/4 = 0.5$.

- Binary vectors $A = [1, 0, 1], B = [1, 0, 1]$: Jaccard = 1.0 (same set of 1s).
- **Use Cases:** Binary data (e.g., user click history: 1=clicked, 0=not), set comparisons (e.g., overlapping keywords in documents), anomaly detection (e.g., fraud patterns).
- **Pros:** Intuitive for categorical/binary data; ignores 0-0 matches (relevant for sparse data).
- **Cons:** Poor for dense numeric data; sensitive to small set sizes.

2.2 Core Distance Metrics

A. Hamming Distance

- **Definition:** Number of positions at which corresponding elements of two **equal-length strings/vectors** differ.

- **Formula:**
$$\text{Hamming Distance}(A, B) = \sum_{i=1}^n |A_i - B_i|$$
 (for binary vectors: counts mismatches).

- **Examples:**
 - Strings "apple" vs "apply": Hamming = 1 (5th character differs).
 - Binary vectors $A = [1, 0, 1], B = [1, 1, 0]$: Hamming = 2 (positions 2 and 3 differ).
- **Use Cases:** Error correction (e.g., DNA sequences, signal processing), binary data comparison (e.g., fingerprint matching), text editing (number of substitutions).
- **Pros:** Simple to compute; ideal for discrete/categorical data of fixed length.
- **Cons:** Requires vectors of equal length; irrelevant for continuous data.

B. Euclidean Distance (L2 Distance)

- **Definition:** Straight-line distance between two points in a Euclidean space. Measures **magnitude difference**.

- **Formula:**
$$\text{Euclidean Distance}(A, B) = \sqrt{\sum_{i=1}^n (A_i - B_i)^2}$$
- **Examples:**
 - 2D points (1,2) vs (4,6): Euclidean = 5 ($\sqrt{[(3^2)+(4^2)]} = 5$).
 - Vectors $A = [1, 2, 3], B = [4, 5, 6]$: Euclidean = $\sqrt{[(3^2)+(3^2)+(3^2)]} = \sqrt{27} \approx 5.196$.

- **Use Cases:** Continuous numeric data (e.g., sensor readings, image pixels), clustering (k-means), geographic data (GPS coordinates).
- **Pros:** Intuitive; preserves geometric relationships in low-dimensional data.
- **Cons:** Sensitive to feature scaling (e.g., weight in kg vs height in cm); poor for high-dimensional data (curse of dimensionality).

C. Variational Distance (Statistical Distance)

- **Definition:** Measures the difference between two **probability distributions** (discrete or continuous).

$$\text{Variational Distance}(P, Q) = \frac{1}{2} \sum_{i=1}^n |P_i - Q_i|$$

- **Formula** (discrete): (range: [0,1], 0=identical distributions).

- **Example:**

- Distributions $P = [0.3, 0.7]$, $Q = [0.6, 0.4]$: Variational Distance = $0.5 * (|0.3 - 0.6| + |0.7 - 0.4|) = 0.3$.

- **Use Cases:** Probability modeling (e.g., comparing language models), hypothesis testing, generative AI (e.g., GANs: measuring distribution similarity).
- **Pros:** Directly compares statistical behavior; interpretable for probabilistic data.
- **Cons:** Only applicable to distributions (sum of P/Q must = 1); not for raw data.

D. Information-Theoretic Distances (e.g., KL Divergence, Jensen-Shannon Distance)

- **KL Divergence (Kullback-Leibler Divergence):**

- Measures how much one probability distribution P diverges from a reference distribution Q .

$$D_{KL}(P|Q) = \sum_{i=1}^n P_i \log \left(\frac{P_i}{Q_i} \right)$$

- Formula: (range: $[0, \infty)$, 0=identical).

- **Use Case:** Model evaluation (e.g., comparing predicted vs true distribution), NLP (language model training).

- **Limitation:** Asymmetric ($D_{KL}(P|Q) \neq D_{KL}(Q|P)$); undefined if $Q_i = 0$ and $P_i > 0$.

- **Jensen-Shannon Distance (JSD):**

- Symmetric variant of KL divergence (range: $[0,1]$).

- Formula: $JSD(P|Q) = \sqrt{\frac{1}{2}D_{KL}(P|M) + \frac{1}{2}D_{KL}(Q|M)}$, where $M = \frac{P+Q}{2}$.
- **Use Case:** Comparing distributions (e.g., text topic distributions, CV feature distributions) where symmetry is required.
- **Pros:** Symmetric; more stable than KL divergence for sparse data.

E. Ranking Distances (e.g., Spearman's Rho, Kendall's Tau, Discounted Cumulative Gain)

- **Spearman's Rho:**
 - Measures the correlation between the **rankings** of two variables (range: [-1,1], 1=perfect agreement).
 - **Use Case:** Comparing user preference rankings (e.g., "top 5 movies" from two users), ordinal data (e.g., survey ratings: 1-5 stars).
- **Kendall's Tau:**
 - Measures the number of **concordant vs discordant pairs** in rankings (range: [-1,1]).
 - **Use Case:** Small datasets (more efficient than Spearman's Rho), e.g., comparing two reviewers' rankings of research papers.
- **Discounted Cumulative Gain (DCG):**
 - Evaluates the quality of a ranked list (e.g., search engine results) by weighting higher-ranked items more heavily.
 - **Use Case:** Recommendation systems, information retrieval (e.g., "how relevant is the top-10 search results?").